FIN 203: Quantitative Finance

Contents

FIN 203: Quantitative Finance **3**
- Professor Philip Howard . . . 3
- Course Description . . . 3
- Course Website & Documents . . . 3
- Course Policies . . . 3
- Prerequisites . . . 3
- Homework . . . 3
- Exams . . . 3
- Grades . . . 3
- Honor Code . . . 4
- Readings . . . 4

Historical Returns **5**
- Lectures . . . 5
  - Investment Primer . . . 5
  - 1 The Investment Environment . . . 5
  - The Investment Process . . . 6
  - Markets Are Competitive . . . 7
  - 2 Asset Classes and Financial Instruments . . . 8
  - Historical Returns . . . 12
  - Risk, Return, and the Historical Record . . . 12
  - Descriptive Statistics . . . 12
- Current Events . . . 12
  - Historical Market Events . . . 12
  - Valuations are Slipping . . . 12
- Exercises . . . 12
  - Market Returns . . . 12
  - Industry Returns . . . 12
  - Market Risk Analysis . . . 13
  - Portfolio Returns . . . 13
  - Security Returns . . . 13
- Practice Problems . . . 13
  - Excel Questions . . . 13
  - Descriptive Statistics - Analytical Questions . . . 13
  - Descriptive Statistics - Computational Questions . . . 13

Portfolio Theory **14**
- Lectures . . . 14
- Current Events . . . 14
- Exercises . . . 14
- Practice Problems . . . 14

CAPM **15**
Lectures . . . 15
Simple Linear Regression . . . 15
Current Events . . . 15
Exercises . . . 15
Practice Problems . . . 15

Factor Models **16**
Lectures . . . 16
Current Events . . . 16
Exercises . . . 16
Practice Problems . . . 16

Binomial Option Models **17**
Lectures . . . 17
Current Events . . . 17
Exercises . . . 17
Practice Problems . . . 17

Monte Carlo Option Models **18**
Lectures . . . 18
Current Events . . . 18
Exercises . . . 18
Practice Problems . . . 18

FIN 203: Quantitative Finance

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Course Description

Welcome to FIN 203: Quantitative Finance. This course will teach quantitative tools for analysis in finance. We will learn descriptive statistics, numerical optimization, regression analysis, decision trees and simulation analysis. Application of these methods to finance will be stressed throughout the course. We will explore historical data, portfolio theory, asset pricing, and option pricing. Finally, we will achieve proficiency in commonly used software to implement our methods for analyses. Specifically, we will focus on the spreadsheet software Excel and simulation software Crystal Ball.

Course Website & Documents

Announcements, assignments, course materials, and grades will be posted to Sakai. You are required to check Sakai daily as it will be an integral part of the course.

Course Policies

I expect all students to be present and prepared for each class. To facilitate this interaction, students are required to have name placards and display them in class. Students will be randomly cold called to perform problems or explain concepts to the class. Lack of preparation will lower your final grade. No phones, tablets, or laptops are allowed during class unless instructed. I will ask you to leave if you use these devices in class. Bring your laptop to every class. All students are required to have Office 365 (available at Software@WFU) installed on Windows 10 (available via WFU's OnTheHub).

Prerequisites

- MTH 111: Calculus I
- MTH 112: Calculus II
- BEM 201: Quantitative Analysis
- ECN 150: Introduction to Economics
- ACC 111: Financial Accounting
- FIN 231: Principles of Finance

Homework

In this course, we will reinforce our lectures with homework assignments. You are expected to complete all problems by their deadline, as they will help facilitate learning the material covered in lectures. Your lowest homework grade will be dropped. I do not accept late assignments.

Exams

Our midterm will be held from 6pm-9pm on Tuesday, March 5th in rooms 155 and 157.

Our final exam will be held from 9am-noon on Friday, May 3rd in room A27/A28.

Grades

Homework will count 20% of your grade. There will be a midterm and final exam, both consisting of 40% of your grade. If you do poorly on the midterm, I will reweight the homework 25%, the midterm 25% and the

final exam 50%, whichever gives you the higher grade. The +/- grading system will divide the 10-point grade ranges into intervals, i.e. 100 to 94.00 A, 93.99 to 90.00 A-, 89.99 to 87.00 B+, 86.99 to 83.00 B, 82.99 to 80 B-, etc.

There will be no make-up midterms. If you miss the midterm for an officially excused absence, the weight of the missed midterm will be distributed to the final exam. If you do not have an official excuse, you will receive a 0 on the midterm.

Honor Code

Integrity is one of Wake Forest University's School of Business core values. Violations of the honor code will result in an F in the course.

Readings

- Chapter 1: Read all sections except section 7
- Chapter 2:
  - Section 2: Read "Treasury Notes and Bonds" and "Corporate Bonds"
  - Section 3: Read "Common Stock as Ownership Shares" and "Characteristics of Common Stock"
  - Section 4: Read "Stock Market Indexes", "Dow Jones Industrial Average", and "The Standard & Poor's 500 Index"
  - Section 5: Options
- Chapter 5:
  - Sections 3 and 4
  - Section 5: Read "Time Series versus Scenario Analysis", "Expected Returns and the Arithmetic Average", "Variance and Standard Deviation", and "The Reward-to-Volatility (Sharpe) Ratio"
  - Section 6
  - Section 7: SKIP the subsections titled "Expected Shortfall", "Lower Partial Standard Deviation", and "Relative Frequency"
  - Section 8
- Chapter 6: All
- Chapter 7: All
- Chapter 8: Sections 1-3
- Chapter 20: Sections 1-3
- Chapter 21: Section 3

Historical Returns

Lectures

Investment Primer

1 The Investment Environment

Real Assets versus Financial Assets The material wealth of a society is ultimately determined by the productive capacity of its economy, that is, the goods and services its members can create. This capacity is a function of the real assets of the economy: the land, buildings, machines, and knowledge that can be used to produce goods and services.

In contrast to real assets are financial assets such as stocks and bonds. Such securities are no more than sheets of paper or, more likely, computer entries, and they do not contribute directly to the productive capacity of the economy. Instead, these assets are the means by which individuals in well-developed economies hold their claims on real assets. Financial assets are claims to the income generated by real assets (or claims on income from the government). If we cannot own our own auto plant (a real asset), we can still buy shares in Ford or Toyota (financial assets) and thereby share in the income derived from the production of automobiles.

Financial Assets It is common to distinguish among three broad types of financial assets: fixed income, equity, and derivatives. Fixed-income or debt securities promise either a fixed stream of income or a stream of income determined by a specified formula. For example, a corporate bond typically would promise that the bondholder will receive a fixed amount of interest each year. Other so-called floating-rate bonds promise payments that depend on current interest rates. For example, a bond may pay an interest rate that is fixed at 2 percentage points above the rate paid on U.S. Treasury bills. Unless the borrower is declared bankrupt, the payments on these securities are either fixed or determined by formula. For this reason, the investment performance of debt securities typically is least closely tied to the financial condition of the issuer.

Nevertheless, fixed-income securities come in a tremendous variety of maturities and payment provisions. At one extreme, the money market refers to debt securities that are short term, highly marketable, and generally of very low risk. Examples of money market securities are U.S. Treasury bills or bank certificates of deposit (CDs). In contrast, the fixed-income capital market includes long-term securities such as Treasury bonds, as well as bonds issued by federal agencies, state and local municipalities, and corporations. These bonds range from very safe in terms of default risk (for example, Treasury securities) to relatively risky (for example, high-yield or "junk" bonds). They also are designed with extremely diverse provisions regarding payments provided to the investor and protection against the bankruptcy of the issuer. We will take a first look at these securities in Chapter 2 and undertake a more detailed analysis of the debt market in Part Four.

Unlike debt securities, common stock, or equity, in a firm represents an ownership share in the corporation. Equityholders are not promised any particular payment. They receive any dividends the firm may pay and have prorated ownership in the real assets of the firm. If the firm is successful, the value of equity will increase; if not, it will decrease. The performance of equity investments, therefore, is tied directly to the success of the firm and its real assets. For this reason, equity investments tend to be riskier than investments in debt securities. Equity markets and equity valuation are the topics of Part Five.

Finally, derivative securities such as options and futures contracts provide payoffs that are determined by the prices of other assets such as bond or stock prices. For example, a call option on a share of Intel stock might turn out to be worthless if Intel's share price remains below a threshold or "exercise" price such as $20 a share, but it can be quite valuable if the stock price rises above that level. 2 Derivative securities are so named because their values derive from the prices of other assets. For example, the value of the call option will depend on the price of Intel stock. Other important derivative securities are futures and swap contracts.

Derivatives have become an integral part of the investment environment. One use of derivatives, perhaps the primary use, is to hedge risks or transfer them to other parties. This is done successfully every day, and the use of these securities for risk management is so commonplace that the multitrillion-dollar market in derivative assets is routinely taken for granted. Derivatives also can be used to take highly speculative positions, however. Every so often, one of these positions blows up, resulting in well-publicized losses of

hundreds of millions of dollars. While these losses attract considerable attention, they are in fact the exception to the more common use of such securities as risk management tools. Derivatives will continue to play an important role in portfolio construction and the financial system. We will return to this topic later in the text.

Financial Markets and the Economy We stated earlier that real assets determine the wealth of an economy, while financial assets merely represent claims on real assets. Nevertheless, financial assets and the markets in which they trade play several crucial roles in developed economies. Financial assets allow us to make the most of the economy's real assets.

Consumption Timing Some individuals in an economy are earning more than they currently wish to spend. Others, for example, retirees, spend more than they currently earn. How can you shift your purchasing power from high-earnings periods to low-earnings periods of life? One way is to "store" your wealth in financial assets. In high-earnings periods, you can invest your savings in financial assets such as stocks and bonds. In low-earnings periods, you can sell these assets to provide funds for your consumption needs. By so doing, you can "shift" your consumption over the course of your lifetime, thereby allocating your consumption to periods that provide the greatest satisfaction. Thus, financial markets allow individuals to separate decisions concerning current consumption from constraints that otherwise would be imposed by current earnings.

Allocation of Risk Virtually all real assets involve some risk. When Ford builds its auto plants, for example, it cannot know for sure what cash flows those plants will generate. Financial markets and the diverse financial instruments traded in those markets allow investors with the greatest taste for risk to bear that risk, while other, less risk-tolerant individuals can, to a greater extent, stay on the sidelines. For example, if Ford raises the funds to build its auto plant by selling both stocks and bonds to the public, the more optimistic or risk-tolerant investors can buy shares of its stock, while the more conservative ones can buy its bonds. Because the bonds promise to provide a fixed payment, the stockholders bear most of the business risk but reap potentially higher rewards. Thus, capital markets allow the risk that is inherent to all investments to be borne by the investors most willing to bear that risk.

This allocation of risk also benefits the firms that need to raise capital to finance their investments. When investors are able to select security types with the risk-return characteristics that best suit their preferences, each security can be sold for the best possible price. This facilitates the process of building the economy's stock of real assets.

The Investment Process

An investor's portfolio is simply his collection of investment assets. Once the portfolio is established, it is updated or "rebalanced" by selling existing securities and using the proceeds to buy new securities, by investing additional funds to increase the overall size of the portfolio, or by selling securities to decrease the size of the portfolio. Investment assets can be categorized into broad asset classes, such as stocks, bonds, real estate, commodities, and so on. Investors make two types of decisions in constructing their portfolios. The asset allocation decision is the choice among these broad asset classes, while the security selection decision is the choice of which particular securities to hold within each asset class.

Asset allocation also includes the decision of how much of one's portfolio to place in safe assets such as bank accounts or money market securities versus in risky assets. Unfortunately, many observers, even those providing financial advice, appear to incorrectly equate saving with safe investing. 4 "Saving" means that you do not spend all of your current income, and therefore can add to your portfolio. You may choose to invest your savings in safe assets, risky assets, or a combination of both.

"Top-down" portfolio construction starts with asset allocation. For example, an individual who currently holds all of his money in a bank account would first decide what proportion of the overall portfolio ought to be moved into stocks, bonds, and so on. In this way, the broad features of the portfolio are established. For example, while the average annual return on the common stock of large firms since 1926 has been better than

11% per year, the average return on U.S. Treasury bills has been less than 4%. On the other hand, stocks are far riskier, with annual returns (as measured by the Standard & Poor's 500 index) that have ranged as low as –46% and as high as 55%. In contrast, T-bills are effectively riskfree: You know what interest rate you will earn when you buy them. Therefore, the decision to allocate your investments to the stock market or to the money market where Treasury bills are traded will have great ramifications for both the risk and the return of your portfolio. A top-down investor first makes this and other crucial asset allocation decisions before turning to the decision of the particular securities to be held in each asset class.

Security analysis involves the valuation of particular securities that might be included in the portfolio. For example, an investor might ask whether Merck or Pfizer is more attractively priced. Both bonds and stocks must be evaluated for investment attractiveness, but valuation is far more difficult for stocks because a stock's performance usually is far more sensitive to the condition of the issuing firm.

In contrast to top-down portfolio management is the "bottom-up" strategy. In this process, the portfolio is constructed from the securities that seem attractively priced without as much concern for the resultant asset allocation. Such a technique can result in unintended bets on one or another sector of the economy. For example, it might turn out that the portfolio ends up with a very heavy representation of firms in one industry, from one part of the country, or with exposure to one source of uncertainty. However, a bottom-up strategy does focus the portfolio on the assets that seem to offer the most attractive investment opportunities.

For example, here is a brief excerpt from the Web site of the Securities and Exchange Commission. "Your 'savings' are usually put into the safest places or products . . . When you 'invest,' you have a greater chance of losing your money than when you 'save.' " This statement is incorrect: Your investment portfolio can be invested in either safe or risky assets, and your savings in any period is simply the difference between your income and consumption.

Markets Are Competitive

Financial markets are highly competitive. Thousands of intelligent and well-backed analysts constantly scour securities markets searching for the best buys. This competition means that we should expect to find few, if any, "free lunches," securities that are so underpriced that they represent obvious bargains. This no-free-lunch proposition has several implications. Let's examine two.

The Risk–Return Trade-Off Investors invest for anticipated future returns, but those returns rarely can be predicted precisely. There will almost always be risk associated with investments. Actual or realized returns will almost always deviate from the expected return anticipated at the start of the investment period. For example, in 1931 (the worst calendar year for the market since 1926), the S&P 500 index fell by 46%. In 1933 (the best year), the index gained 55%. You can be sure that investors did not anticipate such extreme performance at the start of either of these years.

Naturally, if all else could be held equal, investors would prefer investments with the highest expected return. 5 However, the no-free-lunch rule tells us that all else cannot be held equal. If you want higher expected returns, you will have to pay a price in terms of accepting higher investment risk. If higher expected return can be achieved without bearing extra risk, there will be a rush to buy the high-return assets, with the result that their prices will be driven up. Individuals considering investing in the asset at the now-higher price will find the investment less attractive: If you buy at a higher price, your expected rate of return (that is, profit per dollar invested) is lower. The asset will be considered attractive and its price will continue to rise until its expected return is no more than commensurate with risk. At this point, investors can anticipate a "fair" return relative to the asset's risk, but no more. Similarly, if returns were independent of risk, there would be a rush to sell high-risk assets. Their prices would fall (and their expected future rates of return rise) until they eventually were attractive enough to be included again in investor portfolios. We conclude that there should be a risk–return trade-off in the securities markets, with higher-risk assets priced to offer higher expected returns than lower-risk assets.

Of course, this discussion leaves several important questions unanswered. How should one measure the risk of an asset? What should be the quantitative trade-off between risk (properly measured) and expected return? One would think that risk would have something to do with the volatility of an asset's returns, but this guess

turns out to be only partly correct. When we mix assets into diversified portfolios, we need to consider the interplay among assets and the effect of diversification on the risk of the entire portfolio.

Diversification means that many assets are held in the portfolio so that the exposure to any particular asset is limited. The effect of diversification on portfolio risk, the implications for the proper measurement of risk, and the risk–return relationship are the topics of Part Two. These topics are the subject of what has come to be known as modern portfolio theory. The development of this theory brought two of its pioneers, Harry Markowitz and William Sharpe, Nobel Prizes.

2 Asset Classes and Financial Instruments

The Bond Market The bond market is composed of longer term borrowing or debt instruments than those that trade in the money market. This market includes Treasury notes and bonds, corporate bonds, municipal bonds, mortgage securities, and federal agency debt. These instruments are sometimes said to comprise the fixed-income capital market, because most of them promise either a fixed stream of income or a stream of income that is determined according to a specific formula. In practice, these formulas can result in a flow of income that is far from fixed. Therefore, the term fixed income is probably not fully appropriate. It is simpler and more straightforward to call these securities either debt instruments or bonds.

Treasury Bills U.S. Treasury bills (T-bills, or just bills, for short) are the most marketable of all money market instruments. T-bills represent the simplest form of borrowing: The government raises money by selling bills to the public. Investors buy the bills at a discount from the stated maturity value. At the bill's maturity, the holder receives from the government a payment equal to the face value of the bill. The difference between the purchase price and ultimate maturity value constitutes the investor's earnings.

T-bills are issued with initial maturities of 4, 13, 26, or 52 weeks. Individuals can purchase T-bills directly, at auction, or on the secondary market from a government securities dealer. T-bills are highly liquid; that is, they are easily converted to cash and sold at low transaction cost and with not much price risk. Unlike most other money market instruments, which sell in minimum denominations of $100,000, T-bills sell in minimum denominations of only $100, although $10,000 denominations are far more common. The income earned on T-bills is exempt from all state and local taxes, another characteristic distinguishing them from other money market instruments.

Figure 2.1 is a partial listing of T-bill rates. Rather than providing prices of each bill, the financial press reports yields based on those prices. You will see yields corresponding to both bid and ask prices. The ask price is the price you would have to pay to buy a T-bill from a securities dealer. The bid price is the slightly lower price you would receive if you wanted to sell a bill to a dealer. The bid–ask spread is the difference in these prices, which is the dealer's source of profit. (Notice in Figure 2.1 that the bid yield is higher than the ask yield. This is because prices and yields are inversely related.)

The first two yields in Figure 2.1 are reported using the bank-discount method. This means that the bill's discount from its maturity or face value is "annualized" based on a 360-day year, and then reported as a percentage of face value. For example, for the highlighted bill maturing on December 20, 2012, days to maturity are 156 and the yield under the column labeled "ASKED" is given as .125%. This means that a dealer was willing to sell the bill at a discount from face value of .125% 3 (156/360) 5 .0542%. So a bill with $10,000 face value could be purchased for $10,000 3 (1 2 .000542) 5 $9,994.58. Similarly, on the basis of the bid yield of .130%, a dealer would be willing to purchase the bill for $10,000 3 (1 2 .00130 3 156/360) 5 $9,994.367. The bank discount method for computing yields has a long tradition, but it is flawed for at least two reasons. First, it assumes that the year has only 360 days. Second, it computes the yield as a fraction of par value rather than of the price the investor paid to acquire the bill. 1 An investor who buys the bill for the ask price and holds it until maturity will see her investment grow over 156 days by a multiple of $10,000/$9,994.58 5 1.000542, for a gain of .0542%. Annualizing this gain using a 365-day year results in a yield of .0542% 3 365/156 5 .127%, which is the value reported in the last column under "ASK YLD." This last value is called the Treasury-bill's bond-equivalent yield.

Treasury Notes and Bonds The U.S. government borrows funds in large part by selling Treasury notes and Treasury bonds . T-notes are issued with maturities ranging up to 10 years, while bonds are issued with

maturities ranging from 10 to 30 years. Both notes and bonds may be issued in increments of $100 but far more commonly trade in denominations of $1,000. Both notes and bonds make semiannual interest payments called coupon payments, a name derived from precomputer days, when investors would literally clip coupons attached to the bond and present a coupon to receive the interest payment. Figure 2.3 is a listing of Treasury issues. Notice the highlighted note that matures in November 2015. Its bid price is 113.5078. (This is the decimal version of 11365/128 . The minimum tick size, or price increment in the Wall Street Journal listing, is generally 1/128 of a point.) Although bonds are typically traded in denominations of $1,000 par value, prices are quoted as a percentage of par. Thus, the bid price should be interpreted as 113.5078% of par, or $1,135.078 for the $1,000 par value bond. Similarly, the ask price at which the bond could be sold to a dealer is 113.5391% of par, or $1,135.391. The 2 .0859 change means that the closing price on this day fell by .0859% of par value (equivalently, by 11/128 of a point) from the previous day's close. Finally, the yield to maturity based on the ask price is .398%. The yield to maturity reported in the financial pages is calculated by determining the semiannual yield and then doubling it, rather than compounding it for two half-year periods. This use of a simple interest technique to annualize means that the yield is quoted on an annual percentage rate (APR) basis rather than as an effective annual yield. The APR method in this context is also called the bond equivalent yield. We discuss the yield to maturity in more detail in Part Four.

Equity Securities

Common Stock as Ownership Shares

Common stocks, also known as equity securities or equities, represent ownership shares in a corporation. Each share of common stock entitles its owner to one vote on any matters of corporate governance that are put to a vote at the corporation's annual meeting and to a share in the financial benefits of ownership.

The corporation is controlled by a board of directors elected by the shareholders. The board, which meets only a few times each year, selects managers who actually run the corporation on a day-to-day basis. Managers have the authority to make most business decisions without the board's specific approval. The board's mandate is to oversee the management to ensure that it acts in the best interests of shareholders. The members of the board are elected at the annual meeting. Shareholders who do not attend the annual meeting can vote by proxy, empowering another party to vote in their name. Management usually solicits the proxies of shareholders and normally gets a vast majority of these proxy votes. Thus, management usually has considerable discretion to run the firm as it sees fit—without daily oversight from the equityholders who actually own the firm.

We noted in Chapter 1 that such separation of ownership and control can give rise to "agency problems," in which managers pursue goals not in the best interests of shareholders. However, there are several mechanisms that alleviate these agency problems. Among these are compensation schemes that link the success of the manager to that of the firm; oversight by the board of directors as well as outsiders such as security analysts, creditors, or large institutional investors; the threat of a proxy contest in which unhappy shareholders attempt to replace the current management team; or the threat of a takeover by another firm. The common stock of most large corporations can be bought or sold freely on one or more stock exchanges. A corporation whose stock is not publicly traded is said to be closely held. In most closely held corporations, the owners of the firm also take an active role in its management. Therefore, takeovers are generally not an issue.

Characteristics of Common Stock

The two most important characteristics of common stock as an investment are its residual claim and limited liability features.

Residual claim means that stockholders are the last in line of all those who have a claim on the assets and income of the corporation. In a liquidation of the firm's assets the shareholders have a claim to what is left after all other claimants such as the tax authorities, employees, suppliers, bondholders, and other creditors have been paid. For a firm not in liquidation, shareholders have claim to the part of operating income left over after interest and taxes have been paid. Management can either pay this residual as cash dividends to shareholders or reinvest it in the business to increase the value of the shares.

Limited liability means that the most shareholders can lose in the event of failure of the corporation is their

original investment. Unlike owners of unincorporated businesses, whose creditors can lay claim to the personal assets of the owner (house, car, furniture), corporate shareholders may at worst have worthless stock. They are not personally liable for the firm's obligations.

Stock and Bond Market Indexes

Stock Market Indexes

The daily performance of the Dow Jones Industrial Average is a staple portion of the evening news report. Although the Dow is the best-known measure of the performance of the stock market, it is only one of several indicators. Other more broadly based indexes are computed and published daily. In addition, several indexes of bond market performance are widely available.

The ever-increasing role of international trade and investments has made indexes of foreign financial markets part of the general news as well. Thus foreign stock exchange indexes such as the Nikkei Average of Tokyo and the Financial Times index of London are fast becoming household names.

Dow Jones Averages

The Dow Jones Industrial Average (DJIA) of 30 large, "blue-chip" corporations has been computed since 1896. Its long history probably accounts for its preeminence in the public mind. (The average covered only 20 stocks until 1928.) Originally, the DJIA was calculated as the average price of the stocks included in the index. Thus, one would add up the prices of the 30 stocks in the index and divide by 30. The percentage change in the DJIA would then be the percentage change in the average price of the 30 shares.

This procedure means that the percentage change in the DJIA measures the return (excluding dividends) on a portfolio that invests one share in each of the 30 stocks in the index. The value of such a portfolio (holding one share of each stock in the index) is the sum of the 30 prices. Because the percentage change in the average of the 30 prices is the same as the percentage change in the sum of the 30 prices, the index and the portfolio have the same percentage change each day.

Because the Dow corresponds to a portfolio that holds one share of each component stock, the investment in each company in that portfolio is proportional to the company's share price. Therefore, the Dow is called a price-weighted average. You might wonder why the DJIA is now (in early 2013) at a level of about 14,000 if it is supposed to be the average price of the 30 stocks in the index. The DJIA no longer equals the average price of the 30 stocks because the averaging procedure is adjusted whenever a stock splits or pays a stock dividend of more than 10%, or when one company in the group of 30 industrial firms is replaced by another. When these events occur, the divisor used to compute the "average price" is adjusted so as to leave the index unaffected by the event. In the same way that the divisor is updated for stock splits, if one firm is dropped from the average and another firm with a different price is added, the divisor has to be updated to leave the average unchanged by the substitution. By 2013, the divisor for the Dow Jones Industrial Average had fallen to a value of about .1302.

Because the Dow Jones averages are based on small numbers of firms, care must be taken to ensure that they are representative of the broad market. As a result, the composition of the average is changed every so often to reflect changes in the economy. Table 2.5 presents the composition of the Dow industrials in 1928 as well as its composition as of mid-2013. The table presents striking evidence of the changes in the U.S. economy in the last 85 years. Many of the "bluest of the blue chip" companies in 1928 no longer exist, and the industries that were the backbone of the economy in 1928 have given way to some that could not have been imagined at the time.

Standard & Poor's Indexes

The Standard & Poor's Composite 500 (S&P 500) stock index represents an improvement over the Dow Jones Averages in two ways. First, it is a more broadly based index of 500 firms. Second, it is a market-value-weighted index . In the case of the firms XYZ and ABC in Example 2.2, the S&P 500 would give ABC five times the weight given to XYZ because the market value of its outstanding equity is five times larger, $500 million versus $100 million.

The S&P 500 is computed by calculating the total market value of the 500 firms in the index and the total market value of those firms on the previous day of trading. The percentage increase in the total market value from one day to the next represents the increase in the index. The rate of return of the index equals the rate of return that would be earned by an investor holding a portfolio of all 500 firms in the index in proportion to their market values, except that the index does not reflect cash dividends paid by those firms. Actually, most indexes today use a modified version of market-value weights. Rather than weighting by total market value, they weight by the market value of free float, that is, by the value of shares that are freely tradable among investors. For example, this procedure does not count shares held by founding families or governments. These shares are effectively not available for investors to purchase. The distinction is more important in Japan and Europe, where a higher fraction of shares are held in such nontraded portfolios.

A nice feature of both market-value-weighted and price-weighted indexes is that they reflect the returns to straightforward portfolio strategies. If one were to buy shares in each component firm in the index in proportion to its outstanding market value, the value-weighted index would perfectly track capital gains on the underlying portfolio. Similarly, a price-weighted index tracks the returns on a portfolio comprised of an equal number of shares of each firm.

Investors today can easily buy market indexes for their portfolios. One way is to purchase shares in mutual funds that hold shares in proportion to their representation in the S&P 500 or another index. These index funds yield a return equal to that of the index and so provide a low-cost passive investment strategy for equity investors. Another approach is to purchase an exchange-traded fund, or ETF, which is a portfolio of shares that can be bought or sold as a unit, just as one can buy or sell a single share of stock. Available ETFs range from portfolios that track extremely broad global market indexes all the way to narrow industry indexes. We discuss both mutual funds and ETFs in detail in Chapter 4. Standard & Poor's also publishes a 400-stock Industrial Index, a 20-stock Transportation Index, a 40-stock Utility Index, and a 40-stock Financial Index.

Other U.S. Market-Value Indexes

The New York Stock Exchange publishes a market-value-weighted composite index of all NYSE-listed stocks, in addition to subindexes for industrial, utility, transportation, and financial stocks. These indexes are even more broadly based than the S&P 500. The National Association of Securities Dealers publishes an index of more than 3,000 firms traded on the NASDAQ market.

The ultimate U.S. equity index so far computed is the Wilshire 5000 index of the market value of essentially all actively traded stocks in the U.S. Despite its name, the index actually includes more than 5,000 stocks. The performance of many of these indexes appears daily in The Wall Street Journal.

Equally Weighted Indexes

Market performance is sometimes measured by an equally weighted average of the returns of each stock in an index. Such an averaging technique, by placing equal weight on each return, corresponds to an implicit portfolio strategy that invests equal dollar values in each stock. This is in contrast to both price weighting (which requires equal numbers of shares of each stock) and market-value weighting (which requires investments in proportion to outstanding value).

Unlike price- or market-value-weighted indexes, equally weighted indexes do not correspond to buy-and-hold portfolio strategies. Suppose that you start with equal dollar investments in the two stocks of Table 2.3 , ABC and XYZ. Because ABC increases in value by 20% over the year while XYZ decreases by 10%, your portfolio no longer is equally weighted. It is now more heavily invested in ABC. To reset the portfolio to equal weights, you would need to rebalance: Sell off some ABC stock and/or purchase more XYZ stock. Such rebalancing would be necessary to align the return on your portfolio with that on the equally weighted index.

2.5 Derivative Markets

One of the most significant developments in financial markets in recent years has been the growth of futures, options, and related derivatives markets. These instruments provide payoffs that depend on the values of other assets such as commodity prices, bond and stock prices, or market index values. For this reason these instruments sometimes are called derivative assets. Their values derive from the values of other assets.

Options

A call option gives its holder the right to purchase an asset for a specified price, called the exercise or strike price , on or before a specified expiration date. For example, a July call option on IBM stock with an exercise price of $180 entitles its owner to purchase IBM stock for a price of $180 at any time up to and including the expiration date in July. Each option contract is for the purchase of 100 shares. However, quotations are made on a pershare basis. The holder of the call need not exercise the option; it will be profitable to exercise only if the market value of the asset that may be purchased exceeds the exercise price. When the market price exceeds the exercise price, the option holder may "call away" the asset for the exercise price and reap a payoff equal to the difference between the stock price and the exercise price. Otherwise, the option will be left unexercised. If not exercised before the expiration date of the contract, the option simply expires and no longer has value. Calls therefore provide greater profits when stock prices increase and thus represent bullish investment vehicles.

In contrast, a put option gives its holder the right to sell an asset for a specified exercise price on or before a specified expiration date. A July put on IBM with an exercise price of $180 thus entitles its owner to sell IBM stock to the put writer at a price of $180 at any time before expiration in July, even if the market price of IBM is lower than $180. Whereas profits on call options increase when the asset increases in value, profits on put options increase when the asset value falls. The put is exercised only if its holder can deliver an asset worth less than the exercise price in return for the exercise price.

Figure 2.10 is an excerpt of the options quotations for IBM from the online edition of The Wall Street Journal. The price of IBM shares on this date was $183.65. The first two columns give the expiration month and exercise (or strike) price for each option. We have included listings for call and put options with exercise prices of $180 and $185 per share and with expiration dates in July, August, and October 2012 and January 2013.

The next columns provide the closing prices, trading volume, and open interest (outstanding contracts) of each option. For example, 1,998 contracts traded on the July 2012 expiration call with an exercise price of $180. The last trade was at $5.50, meaning that an option to purchase one share of IBM at an exercise price of $180 sold for $5.50. Each option contract (on 100 shares) therefore costs $550.

Notice that the prices of call options decrease as the exercise price increases. For example, the July expiration call with exercise price $185 costs only $2.80. This makes sense, because the right to purchase a share at a higher price is less valuable. Conversely, put prices increase with the exercise price. The right to sell IBM at a price of $180 in July costs $2.11, while the right to sell at $185 costs $4.20.

Option prices also increase with time until expiration. Clearly, one would rather have the right to buy IBM for $180 at any time until October rather than at any time until July. Not surprisingly, this shows up in a higher price for the October expiration options. For example, the call with exercise price $180 expiring in October sells for $9.70 compared to only $5.50 for the July call.

Historical Returns

Risk, Return, and the Historical Record

Descriptive Statistics

Current Events

Historical Market Events

Valuations are Slipping

Exercises

Market Returns

Industry Returns

Description Attached are historical US industry portfolio returns The objective of this exercise is to explore the risk-reward relationship across US industries.

Directions

1. Calculate the expected return for each industry portfolio.
2. Calculate the standard deviation for each industry portfolio.
3. Calculate the correlation between expected returns and standard deviations.
4. Replicate the plot below by plotting the expected returns against the standard deviations.

Questions

- In finance, we will interpret average returns as a reward and standard deviations as a risk of investing. Looking at your plot, is there a risk-reward relationship? If so, describe it.
- What does the correlation coefficient calculated above communicate to us?

Solutions

Market Risk Analysis

Portfolio Returns

Security Returns

Practice Problems

Excel Questions

Descriptive Statistics - Analytical Questions

Descriptive Statistics - Computational Questions

Portfolio Theory

Lectures

Current Events

Exercises

Practice Problems

CAPM

Lectures

Simple Linear Regression

In statistics, simple linear regression is a linear regression model with a single explanatory variable.[1][2][3][4][5] That is, it concerns two-dimensional sample points with one independent variable and one dependent variable (conventionally, the x and y coordinates in a Cartesian coordinate system) and finds a linear function (a non-vertical straight line) that, as accurately as possible, predicts the dependent variable values as a function of the independent variables. The adjective simple refers to the fact that the outcome variable is related to a single predictor.

It is common to make the additional stipulation that the ordinary least squares (OLS) method should be used: the accuracy of each predicted value is measured by its squared residual (vertical distance between the point of the data set and the fitted line), and the goal is to make the sum of these squared deviations as small as possible. Other regression methods that can be used in place of ordinary least squares include least absolute deviations (minimizing the sum of absolute values of residuals) and the Theil–Sen estimator (which chooses a line whose slope is the median of the slopes determined by pairs of sample points). Deming regression (total least squares) also finds a line that fits a set of two-dimensional sample points, but (unlike ordinary least squares, least absolute deviations, and median slope regression) it is not really an instance of simple linear regression, because it does not separate the coordinates into one dependent and one independent variable and could potentially return a vertical line as its fit.

Current Events

Exercises

Practice Problems

Factor Models

Lectures

Current Events

Exercises

Practice Problems

Binomial Option Models

Lectures

Current Events

Exercises

Practice Problems

Monte Carlo Option Models

Lectures

Current Events

Exercises

Practice Problems